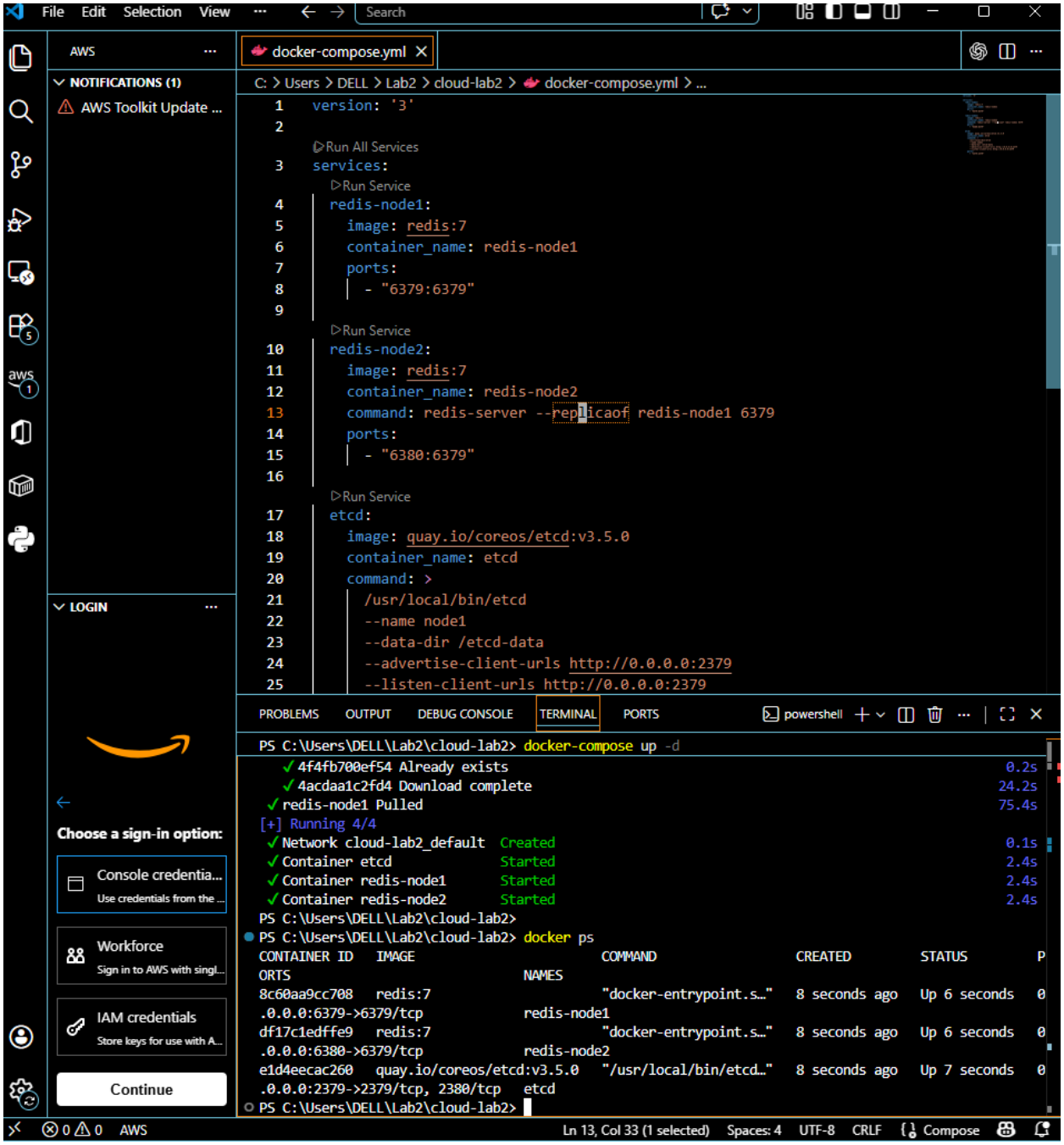
Lab2

Terminal Outputs:



File Edit Selection ··· Search Update

AWS

NOTIFICATIONS (1)

AWS Toolkit Update ...

docker-compose.yml

C: > Users > DELL > Lab2 > cloud-lab2 > docker-compose.yml > ...

```yaml
version: '3'

services:
  redis-node1:
    image: redis:7
    container_name: redis-node1
    ports:
      - "6379:6379"

  redis-node2:
    image: redis:7
    container_name: redis-node2
    command: redis-server --replicaof redis-node1 6379
    ports:
      - "6380:6379"

  etcd:
    image: quay.io/coreos/etcd:v3.5.0
    container_name: etcd
    command: >
      /usr/local/bin/etcd
      --name node1
      --data-dir /etcd-data
      --advertise-client-urls http://0.0.0.0:2379
      --listen-client-urls http://0.0.0.0:2379
```

LOGIN

Choose a sign-in option:

Console credentia...
Use credentials from the ...

Workforce
Sign in to AWS with singl...

IAM credentials
Store keys for use with A...

Continue

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
PS C:\Users\DELL\Lab2\cloud-lab2> docker ps
nds   0.0.0.0:6379->6379/tcp            redis-node1
df17c1edffe9   redis:7                    "docker-entrypoint.s…"   8 seconds ago   Up 6 seco
nds   0.0.0.0:6380->6379/tcp            redis-node2
e1d4eecac260   quay.io/coreos/etcd:v3.5.0   "/usr/local/bin/etcd…"   8 seconds ago   Up 7 seco
nds   0.0.0.0:2379->2379/tcp, 2380/tcp   etcd
PS C:\Users\DELL\Lab2\cloud-lab2>
PS C:\Users\DELL\Lab2\cloud-lab2> docker exec -it redis-node1 redis-cli
127.0.0.1:6379>
127.0.0.1:6379> SET key value [NX|XX] [GET] [EX seconds|PX milliseconds|EXAT unix-time-seconds
127.0.0.1:6379> SET u key value [NX|XX] [GET] [EX seconds|PX milliseconds|EXAT unix-time-secon
127.0.0.1:6379> SET us key value [NX|XX] [GET] [EX seconds|PX milliseconds|EXAT unix-time-seco
127.0.0.1:6379> SET use key value [NX|XX] [GET] [EX seconds|PX milliseconds|EXAT unix-time-sec
127.0.0.1:6379> SET user key value [NX|XX] [GET] [EX seconds|PX milliseconds|EXAT unix-time-se
127.0.0.1:6379> SET user value [NX|XX] [GET] [EX seconds|PX milliseconds|EXAT unix-time-second
127.0.0.1:6379> SET user "Yousef" value [NX|XX] [GET] [EX seconds|PX milliseconds|EXAT unix-ti
me-seco
PS C:\Users\DELL\Lab2\cloud-lab2> SET user "Yousef"
PS C:\Users\DELL\Lab2\cloud-lab2>
PS C:\Users\DELL\Lab2\cloud-lab2>
```

0 0 AWS Ln 13, Col 33 (1 selected) Spaces: 4 UTF-8 CRLF Compose

File Edit Selection ··· Search Update

AWS ···

NOTIFICATIONS (1)

AWS Toolkit Update ...

LOGIN ···

Choose a sign-in option:

Console credentia... Use credentials from the ...

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IAM Workforce Store keys for use with A...

Continue

docker-compose.yml

C: > Users > DELL > Lab2 > cloud-lab2 > docker-compose.yml > ...

```yaml
version: '3'

services:
  redis-node1:
    image: redis:7
    container_name: redis-node1
    ports:
      - "6379:6379"

  redis-node2:
    image: redis:7
    container_name: redis-node2
    command: redis-server --replicaof redis-node1 6379
    ports:
      - "6380:6379"

  etcd:
    image: quay.io/coreos/etcd:v3.5.0
    container_name: etcd
    command: >
      /usr/local/bin/etcd
      --name node1
      --data-dir /etcd-data
      --advertise-client-urls http://0.0.0.0:2379
      --listen-client-urls http://0.0.0.0:2379
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS docker

```
PS C:\Users\DELL> cd C:\Users\DELL\Lab2\cloud-lab2
PS C:\Users\DELL\Lab2\cloud-lab2>
PS C:\Users\DELL\Lab2\cloud-lab2> docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET user
(nil)
127.0.0.1:6379>
```

0 0 AWS Ln 13, Col 33 (1 selected) Spaces: 4 UTF-8 CRLF Compose

File Edit Selection ... Search Update

AWS

NOTIFICATIONS (1)

AWS Toolkit Update ...

docker-compose.yml

C: > Users > DELL > Lab2 > cloud-lab2 > docker-compose.yml > ...

```yaml
version: '3'

services:
  redis-node1:
    image: redis:7
    container_name: redis-node1
    ports:
      - "6379:6379"

  redis-node2:
    image: redis:7
    container_name: redis-node2
    command: redis-server --replicaof redis-node1 6379
    ports:
      - "6380:6379"

  etcd:
    image: quay.io/coreos/etcd:v3.5.0
    container_name: etcd
    command: >
      /usr/local/bin/etcd
      --name node1
      --data-dir /etcd-data
      --advertise-client-urls http://0.0.0.0:2379
      --listen-client-urls http://0.0.0.0:2379
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS docker

```
PS C:\Users\DELL\Lab2\cloud-lab2>
PS C:\Users\DELL\Lab2\cloud-lab2> docker stop redis-node2
redis-node2
PS C:\Users\DELL\Lab2\cloud-lab2> SET age 22
PS C:\Users\DELL\Lab2\cloud-lab2> docker start redis-node2
redis-node2
PS C:\Users\DELL\Lab2\cloud-lab2> GET age
GET : The term 'GET' is not recognized as the name of a cmdlet, function, script file, or
operable program. Check the spelling of the name, or if a path was included, verify that the
path is correct and try again.
At line:1 char:1
+ GET age
+ ~~~
    + CategoryInfo          : ObjectNotFound: (GET:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\DELL\Lab2\cloud-lab2> docker exec -it redis-node1 redis-cli
127.0.0.1:6379> SET user "Yousef"
OK
127.0.0.1:6379>
```

0 0 AWS Ln 13, Col 33 (1 selected) Spaces: 4 UTF-8 CRLF Compose

File Edit Selection ... Search Update

AWS

NOTIFICATIONS (1)

AWS Toolkit Update ...

docker-compose.yml

C: > Users > DELL > Lab2 > cloud-lab2 > docker-compose.yml > ...

```yaml
version: '3'

services:
  redis-node1:
    image: redis:7
    container_name: redis-node1
    ports:
      - "6379:6379"

  redis-node2:
    image: redis:7
    container_name: redis-node2
    command: redis-server --replicaof redis-node1 6379
    ports:
      - "6380:6379"

  etcd:
    image: quay.io/coreos/etcd:v3.5.0
    container_name: etcd
    command: >
      /usr/local/bin/etcd
      --name node1
      --data-dir /etcd-data
      --advertise-client-urls http://0.0.0.0:2379
      --listen-client-urls http://0.0.0.0:2379
```

LOGIN

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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS docker

```
PS C:\Users\DELL\Lab2\cloud-lab2> GET age
    + CategoryInfo          : ObjectNotFound: (GET:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\DELL\Lab2\cloud-lab2> docker start redis-node2
redis-node2
PS C:\Users\DELL\Lab2\cloud-lab2> GET age
GET : The term 'GET' is not recognized as the name of a cmdlet, function, script file, or
operable program. Check the spelling of the name, or if a path was included, verify that the
path is correct and try again.
At line:1 char:1
+ GET age
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    + CategoryInfo          : ObjectNotFound: (GET:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\DELL\Lab2\cloud-lab2> docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET user
"Yousef"
127.0.0.1:6379>
```

0 0 AWS Ln 13, Col 33 (1 selected) Spaces: 4 UTF-8 CRLF Compose

docker-compose.yml

C: > Users > DELL > Lab2 > cloud-lab2 > docker-compose.yml > ...

```yaml
version: '3'

services:
  redis-node1:
    image: redis:7
    container_name: redis-node1
    ports:
      - "6379:6379"

  redis-node2:
    image: redis:7
    container_name: redis-node2
    command: redis-server --replicaof redis-node1 6379
    ports:
      - "6380:6379"

  etcd:
    image: quay.io/coreos/etcd:v3.5.0
    container_name: etcd
    command: >
      /usr/local/bin/etcd
```

https://quay.io/repository/coreos/etcd (ctrl + click)

C:\WINDOWS\system32\cmd.

```
C:\Users\DELL\Lab2\cloud-lab2>docker-compose up -d
empty compose file

C:\Users\DELL\Lab2\cloud-lab2>docker stop redis-node2
redis-node2

C:\Users\DELL\Lab2\cloud-lab2>docker start redis-node2
redis-node2

C:\Users\DELL\Lab2\cloud-lab2>
```

```
PS C:\Users\DELL\Lab2\cloud-lab2> docker exec -it redis-node1 redis-cli
127.0.0.1:6379> SET user "Yousef"
OK
127.0.0.1:6379> SET age 22
OK
127.0.0.1:6379> SET age 22
OK
127.0.0.1:6379> GET age
"22\x16\x16"
127.0.0.1:6379>
```

```
C:\WINDOWS\system32\cmd.

C:\Users\DELL\Lab2\cloud-lab2>docker exec -it etcd etcdctl put foo
 bar
OK

C:\Users\DELL\Lab2\cloud-lab2>docker exec -it etcd etcdctl get foo

foo
bar

C:\Users\DELL\Lab2\cloud-lab2>
```

```
C:\WINDOWS\system32\cmd.
C:\Users\DELL\Lab2\cloud-lab2>docker exec -it etcd etcdctl endpoint status --write-ou
t=table
+----------------+------------------+---------+---------+-----------+------------+---
--------+------------+--------------------+--------+
|    ENDPOINT    |        ID        | VERSION | DB SIZE | IS LEADER | IS LEARNER | RA
FT TERM | RAFT INDEX | RAFT APPLIED INDEX | ERRORS |
+----------------+------------------+---------+---------+-----------+------------+---
--------+------------+--------------------+--------+
| 127.0.0.1:2379 | 8e9e05c52164694d |   3.5.0 |   20 kB |      true |      false |
      2 |          5 |                  5 |        |
+----------------+------------------+---------+---------+-----------+------------+---
--------+------------+--------------------+--------+

C:\Users\DELL\Lab2\cloud-lab2>
```

NOTIFICATIONS (1)

AWS Toolkit Update ...

C: > Users > DELL > Lab2 > cloud-lab2 > docker-compose.yml > ...

```yaml
version: '3'

services:
  redis-node1:
    image: redis:7
    container_name: redis-node1
    ports:
      - "6379:6379"
```

C:\WINDOWS\system32\cmd.

```
C:\Users\DELL\Lab2\cloud-lab2>docker stop etcd
etcd

C:\Users\DELL\Lab2\cloud-lab2>docker start etcd
etcd

C:\Users\DELL\Lab2\cloud-lab2>docker exec -it etcd etcdctl endpoint status --write-ou
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+----------------+------------------+---------+---------+-----------+------------+---
--------+------------+--------------------+--------+
|    ENDPOINT    |        ID        | VERSION | DB SIZE | IS LEADER | IS LEARNER | RA
FT TERM | RAFT INDEX | RAFT APPLIED INDEX | ERRORS |
+----------------+------------------+---------+---------+-----------+------------+---
--------+------------+--------------------+--------+
| 127.0.0.1:2379 | 8e9e05c52164694d |   3.5.0 |   20 kB |      true |      false |
      3 |          7 |                  7 |        |
+----------------+------------------+---------+---------+-----------+------------+---
--------+------------+--------------------+--------+

C:\Users\DELL\Lab2\cloud-lab2>
```

hoose a sign-in option:

Console credentia...
Use credentials from the ...

Workforce
Sign in to AWS with singl...

IAM credentials
Store keys for use with A...

```
At line:1 char:1
+ GET age
+ ~~~~
    + CategoryInfo          : ObjectNotFound: (GET:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\DELL\Lab2\cloud-lab2> docker exec -it redis-node1 redis-cli
127.0.0.1:6379> SET user "Yousef"
OK
127.0.0.1:6379> SET age 22
OK
127.0.0.1:6379> SET age 22
OK
```

The Docker Compose,yaml

```yaml
version: '3'

services:
  redis-node1:
    image: redis:7
    container_name: redis-node1
    ports:
      - "6379:6379"

  redis-node2:
    image: redis:7
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    command: redis-server --replicaof redis-node1 6379
    ports:
      - "6380:6379"

  etcd:
    image: quay.io/coreos/etcd:v3.5.0
    container_name: etcd
    command: >
      /usr/local/bin/etcd
      --name node1
      --data-dir /etcd-data
      --advertise-client-urls http://0.0.0.0:2379
      --listen-client-urls http://0.0.0.0:2379
    ports:
      - "2379:2379"
```

🔹 Redis & CAP Theorem

Redis replication demonstrates eventual consistency because updates from the master node are asynchronously propagated to replicas. During network partition, the system continues to accept writes, prioritizing availability over consistency. Once the partition is resolved, replicas synchronize data.

🔹 etcd & Raft Consensus

etcd uses the Raft consensus algorithm where a leader node is responsible for handling client requests. When the leader fails, a new leader is elected automatically by the remaining nodes. This ensures high availability and strong consistency.